02 - Heat Equation Experiments

AC-PINN Project | Authors: Suyash Vasal Jain, Nishita Raghvendra

PDE: $u_t = \alpha u_{xx}$, $\alpha = 0.01$

IC: $u(x,0) = \sin(\pi x)$ | **BC:** $u(-1,t) = u(1,t) = 0$

Domain: $x \in [-1,1]$, $t \in [0,1]$

Architecture: `[2, 32, 32, 32, 1]` | **Epochs:** 10000

Note: Heat equation is smooth - used as baseline to confirm AC-PINN does not hurt performance on easy problems.

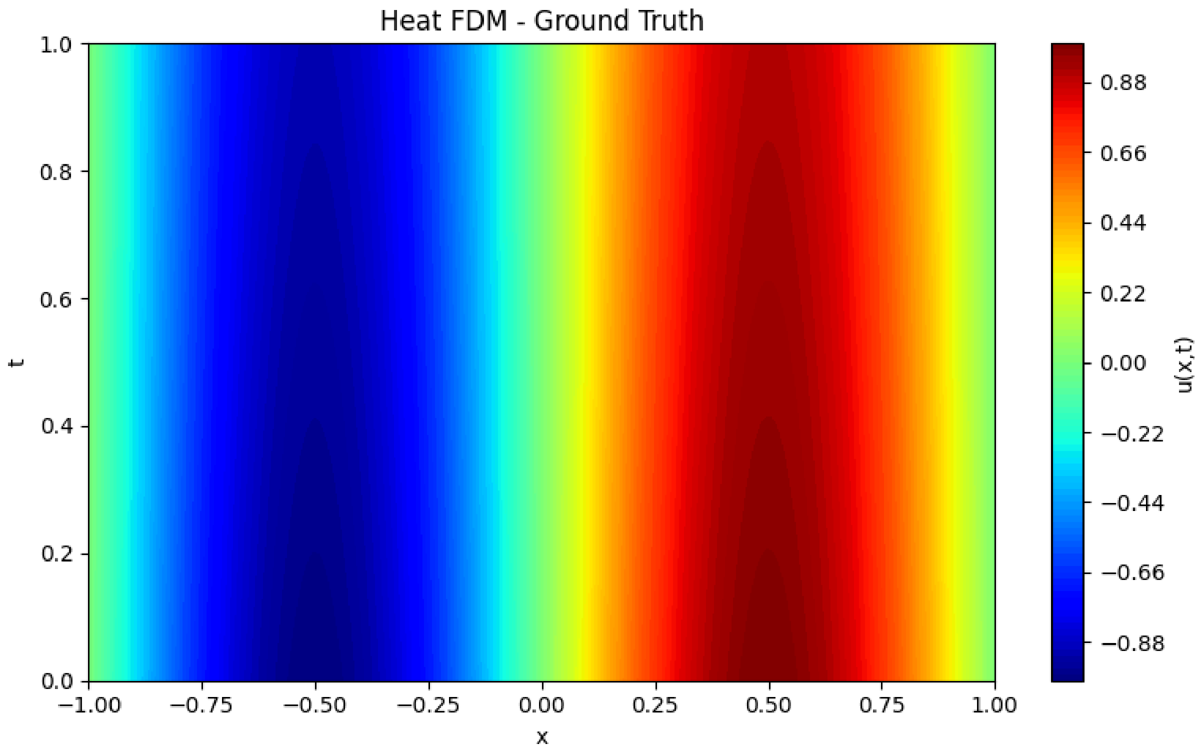
In [1]:
```
import sys, os
sys.path.append('..')
import torch
import numpy as np
import matplotlib.pyplot as plt
from pinn_base import (
    device, NoisyDataGenerator, PINNSolver, ACPINNSolver,
    HeatFDM, Benchmark, save_metrics, save_history, save_training_plots
)

PDE        = 'heat'
LAYERS     = [2, 32, 32, 32, 1]
EPOCHS     = 10000
PDE_PARAMS = {'alpha': 0.01}
RESULTS    = '../results/heat/'
FIGURES    = '../figures/heat/'
os.makedirs(RESULTS, exist_ok=True)
os.makedirs(FIGURES, exist_ok=True)

gen = NoisyDataGenerator(pde=PDE, alpha=0.01)
print(f'Device: {device}')
```

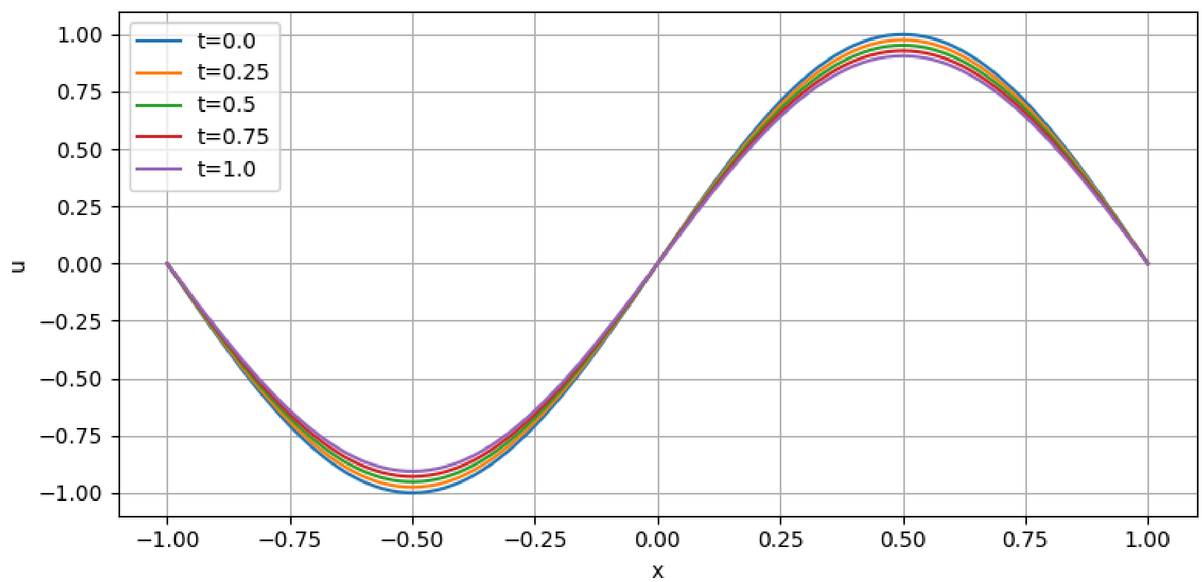
```
Device: cuda
```

Section 1 - FDM Ground Truth

In [2]:
```
fdm = HeatFDM(nx=256, nt=1000, alpha=0.01)
fdm.solve()
fdm.plot_solution(title='Heat FDM - Ground Truth')
fdm.plot_time_slices()
```

```
HeatFDM solved in 0.3426s
```





Section 2 - Data Conditions

In [3]:
```
data_clean_dense  = gen.generate(N_ic=1000, N_bc=1000, N_f=8000, noise_eps=0.0)
data_noisy_sparse = gen.generate(N_ic=20,   N_bc=20,   N_f=2000, noise_eps=0.1)
print('Data ready')
```

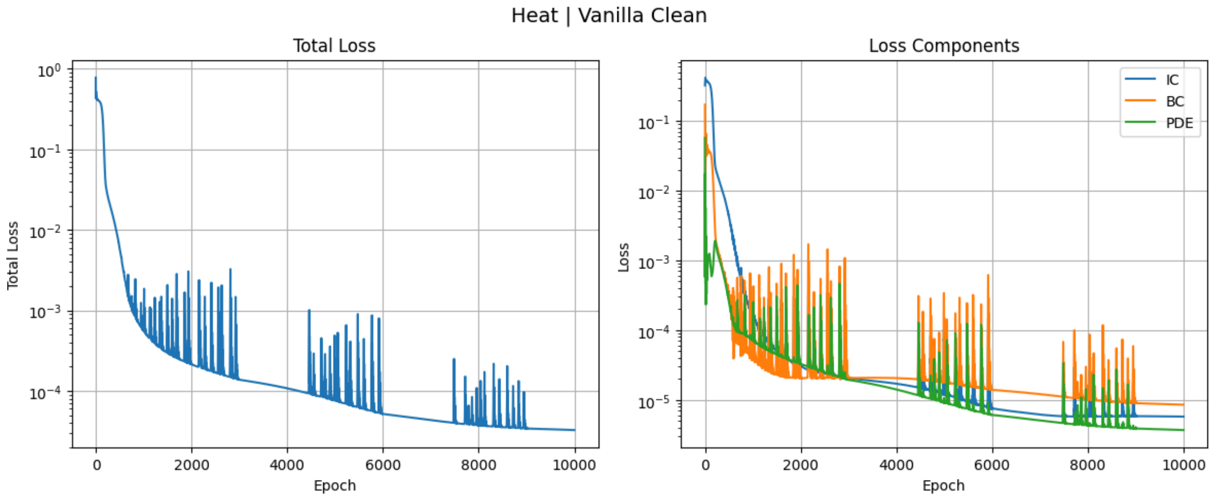
```
Data ready
```

Section 3 - Vanilla PINN, Clean Dense

In [4]:
```
vanilla_clean = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
h_vc = vanilla_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='
```

```
save_training_plots(h_vc, FIGURES+'vanilla_clean_training.png', 'Heat | Vanilla Cle
vanilla_clean.plot_loss_history(h_vc)
vanilla_clean.plot_solution(title='Vanilla PINN - Clean Dense')
vanilla_clean.plot_initial_condition_comparison(gen)
save_history(h_vc, RESULTS+'vanilla_clean_history.npy')
```

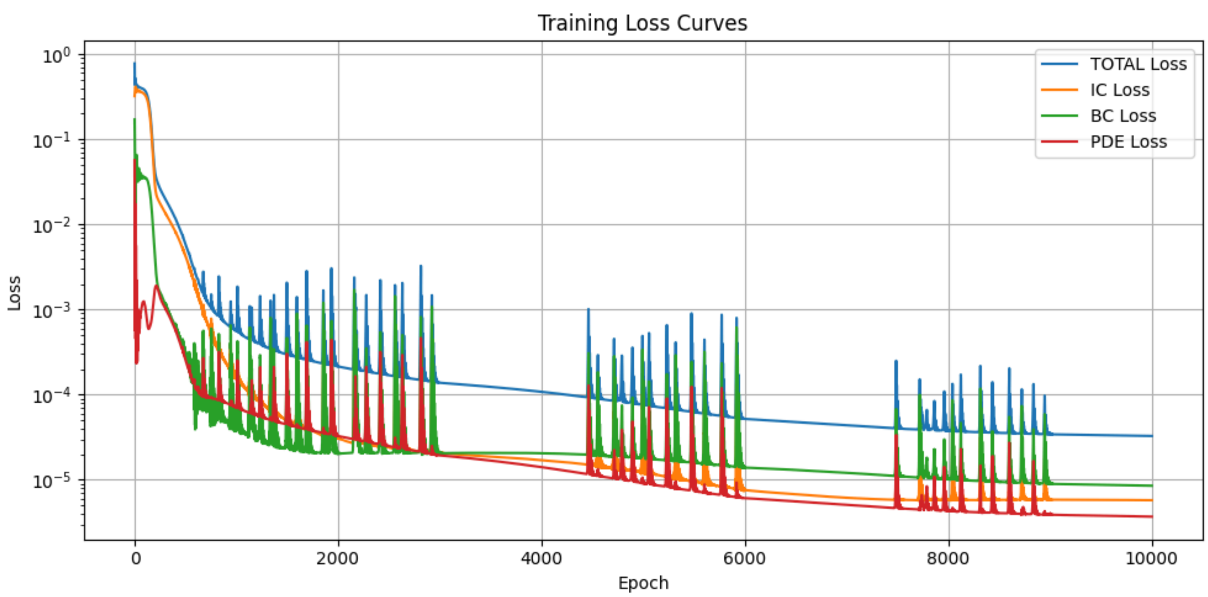
```
D:\PINN\ac-pinn-project\ac-pinn-project\venv\Lib\site-packages\torch\autograd\graph.
py:825: UserWarning: Attempting to run cuBLAS, but there was no current CUDA contex
t! Attempting to set the primary context... (Triggered internally at C:\actions-runn
er\_work\pytorch\pytorch\builder\windows\pytorch\aten\src\ATen\cuda\CublasHandlePoo
l.cpp:135.)
  return Variable._execution_engine.run_backward(  # Calls into the C++ engine to ru
n the backward pass
[Heat | Vanilla Clean | Epoch     0/10000] Total: 0.775922 | IC: 0.321350 | BC: 0.17
0544 | PDE: 0.056806
[Heat | Vanilla Clean | Epoch  1000/10000] Total: 0.000572 | IC: 0.000164 | BC: 0.00
0039 | PDE: 0.000074
[Heat | Vanilla Clean | Epoch  2000/10000] Total: 0.000215 | IC: 0.000031 | BC: 0.00
0020 | PDE: 0.000033
[Heat | Vanilla Clean | Epoch  3000/10000] Total: 0.000138 | IC: 0.000021 | BC: 0.00
0021 | PDE: 0.000019
[Heat | Vanilla Clean | Epoch  4000/10000] Total: 0.000108 | IC: 0.000017 | BC: 0.00
0021 | PDE: 0.000014
[Heat | Vanilla Clean | Epoch  5000/10000] Total: 0.000232 | IC: 0.000035 | BC: 0.00
0149 | PDE: 0.000010
[Heat | Vanilla Clean | Epoch  6000/10000] Total: 0.000052 | IC: 0.000008 | BC: 0.00
0014 | PDE: 0.000006
[Heat | Vanilla Clean | Epoch  7000/10000] Total: 0.000044 | IC: 0.000006 | BC: 0.00
0012 | PDE: 0.000005
[Heat | Vanilla Clean | Epoch  8000/10000] Total: 0.000038 | IC: 0.000006 | BC: 0.00
0010 | PDE: 0.000004
[Heat | Vanilla Clean | Epoch  9000/10000] Total: 0.000035 | IC: 0.000006 | BC: 0.00
0009 | PDE: 0.000004
Training complete in 94.06s
```

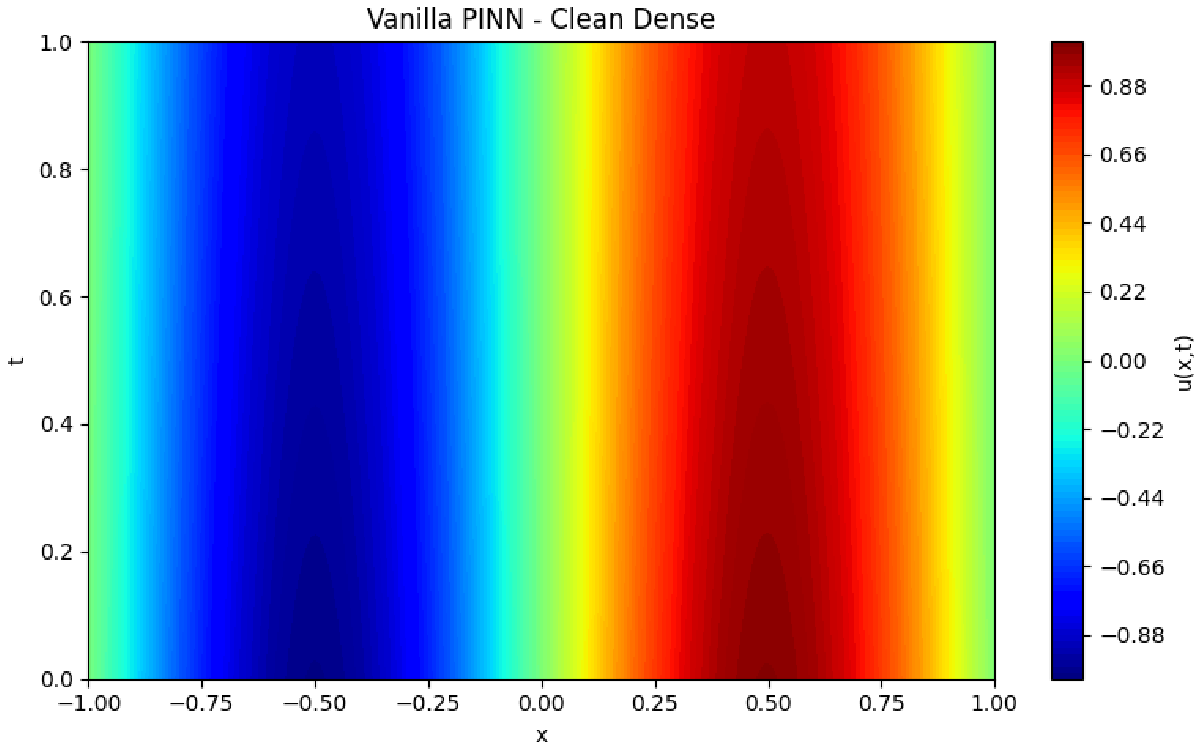


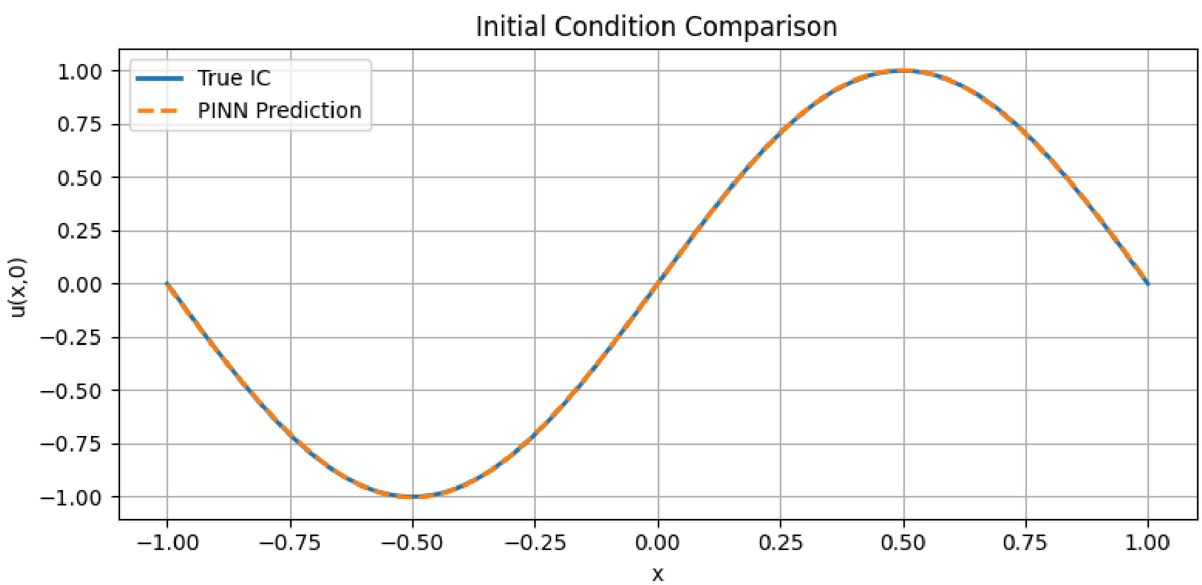
N/A - Vanilla PINN

N/A - Vanilla PINN

```
Saved: ../figures/heat/vanilla_clean_training.png
Saved: ../figures/heat/vanilla_clean_training.csv
```








```
Saved: ../results/heat/vanilla_clean_history.npy
```

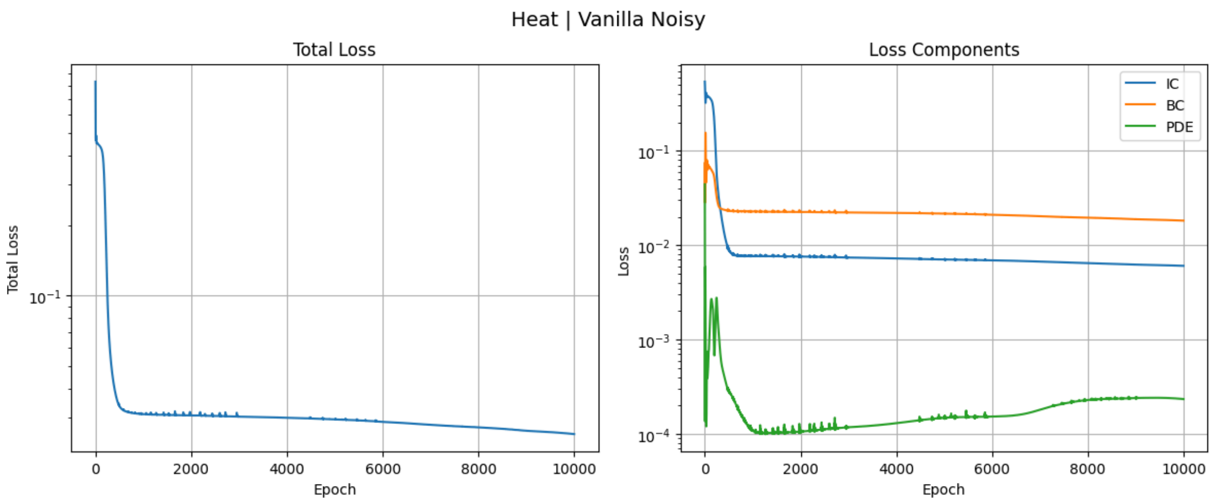
Section 4 - Vanilla PINN, Noisy Sparse

In [5]:

```
vanilla_noisy = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
h_vn = vanilla_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label=
save_training_plots(h_vn, FIGURES+'vanilla_noisy_training.png', 'Heat | Vanilla Noi
vanilla_noisy.plot_loss_history(h_vn)
vanilla_noisy.plot_solution(title='Vanilla PINN - Noisy Sparse')
save_history(h_vn, RESULTS+'vanilla_noisy_history.npy')
```

```
[Heat | Vanilla Noisy | Epoch     0/10000] Total: 0.833671 | IC: 0.539214 | BC: 0.07
3767 | PDE: 0.044138
```

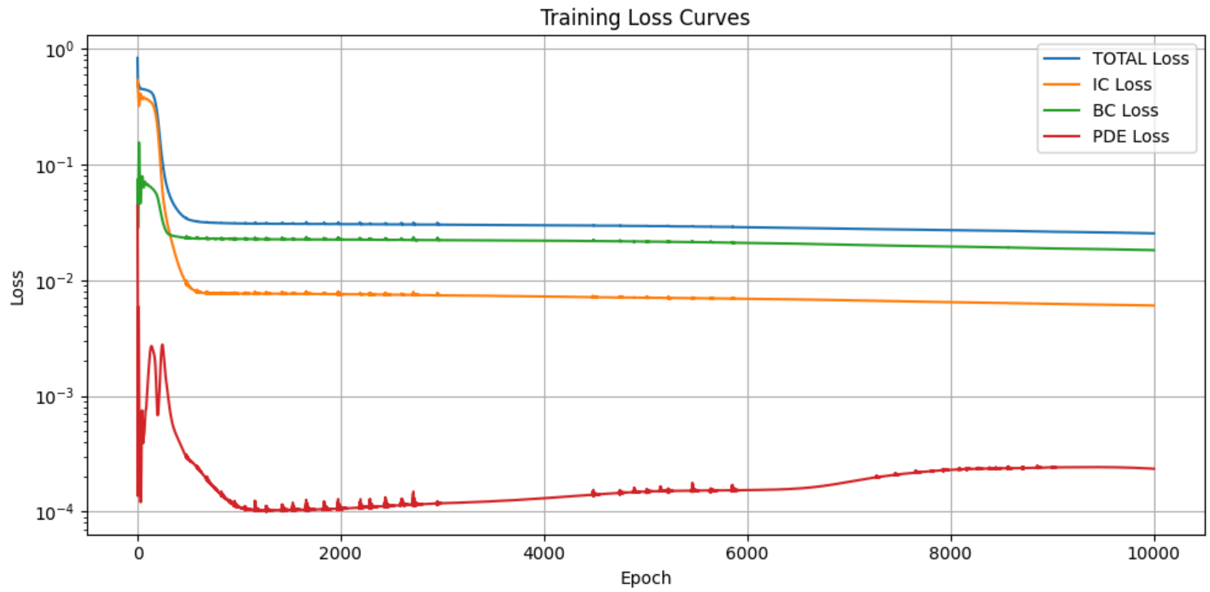
```
[Heat | Vanilla Noisy | Epoch  1000/10000] Total: 0.030958 | IC: 0.007662 | BC: 0.02
2752 | PDE: 0.000109
[Heat | Vanilla Noisy | Epoch  2000/10000] Total: 0.030613 | IC: 0.007621 | BC: 0.02
2451 | PDE: 0.000108
[Heat | Vanilla Noisy | Epoch  3000/10000] Total: 0.030208 | IC: 0.007398 | BC: 0.02
2219 | PDE: 0.000118
[Heat | Vanilla Noisy | Epoch  4000/10000] Total: 0.029883 | IC: 0.007237 | BC: 0.02
1993 | PDE: 0.000130
[Heat | Vanilla Noisy | Epoch  5000/10000] Total: 0.029405 | IC: 0.007052 | BC: 0.02
1613 | PDE: 0.000148
[Heat | Vanilla Noisy | Epoch  6000/10000] Total: 0.028676 | IC: 0.006908 | BC: 0.02
1002 | PDE: 0.000153
[Heat | Vanilla Noisy | Epoch  7000/10000] Total: 0.027862 | IC: 0.006706 | BC: 0.02
0252 | PDE: 0.000181
[Heat | Vanilla Noisy | Epoch  8000/10000] Total: 0.027170 | IC: 0.006463 | BC: 0.01
9560 | PDE: 0.000229
[Heat | Vanilla Noisy | Epoch  9000/10000] Total: 0.026284 | IC: 0.006234 | BC: 0.01
8860 | PDE: 0.000238
Training complete in 93.45s
```

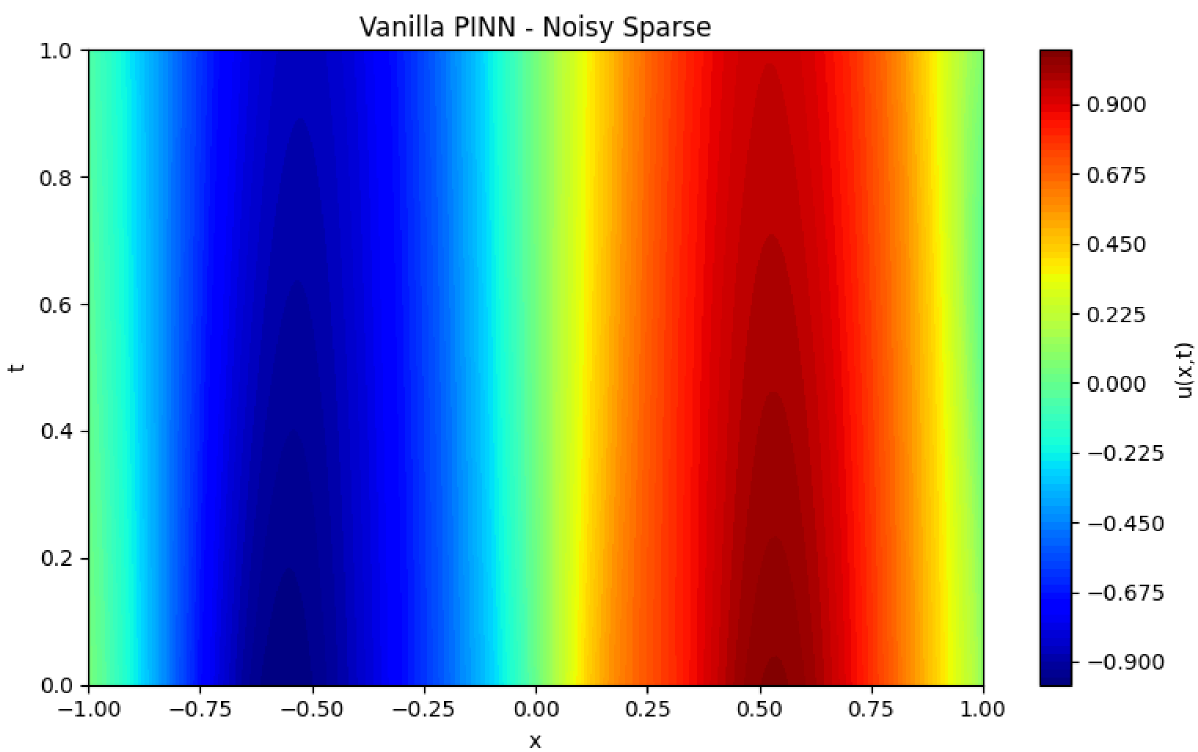


N/A - Vanilla PINN

N/A - Vanilla PINN

```
Saved: ../figures/heat/vanilla_noisy_training.png
Saved: ../figures/heat/vanilla_noisy_training.csv
```





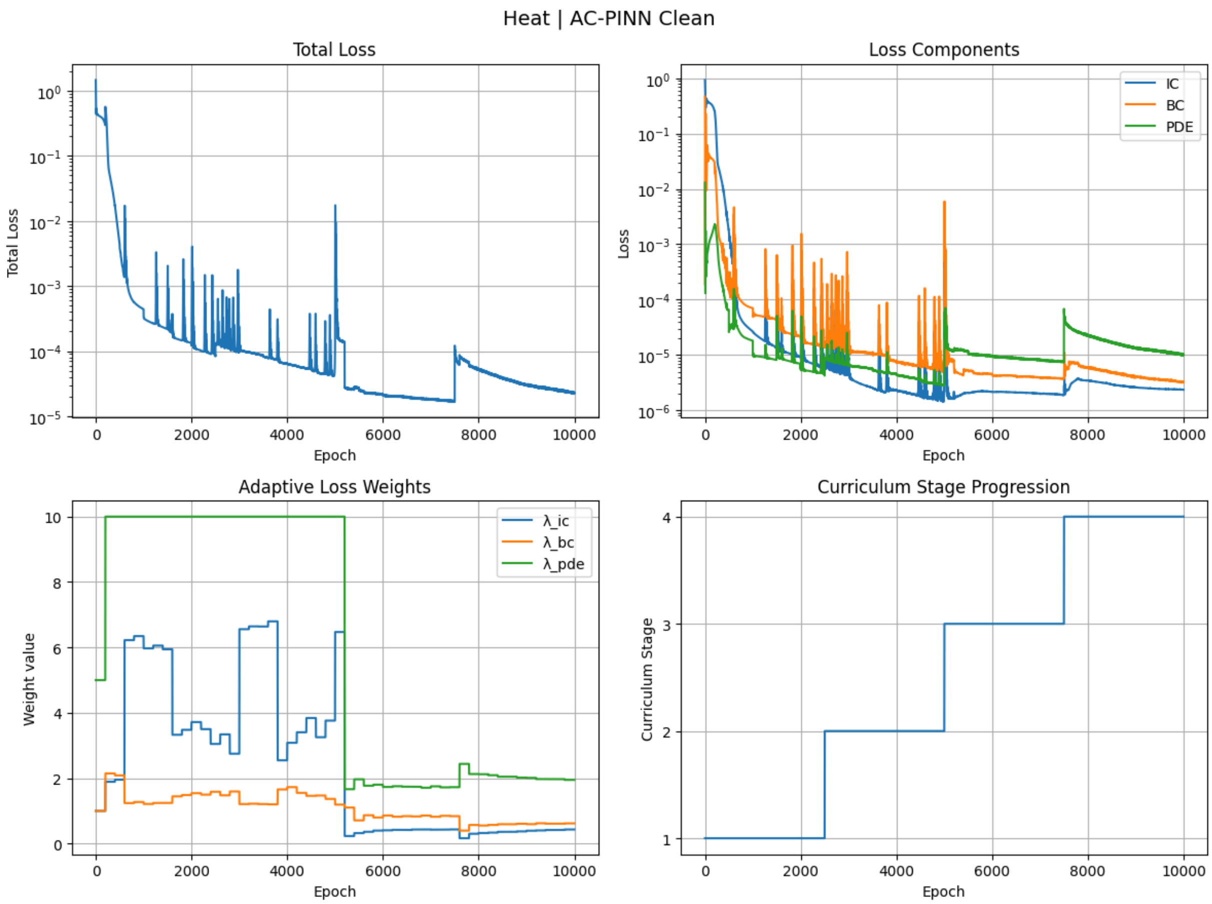

```
Saved: ../results/heat/vanilla_noisy_history.npy
```

Section 5 - AC-PINN, Clean Dense

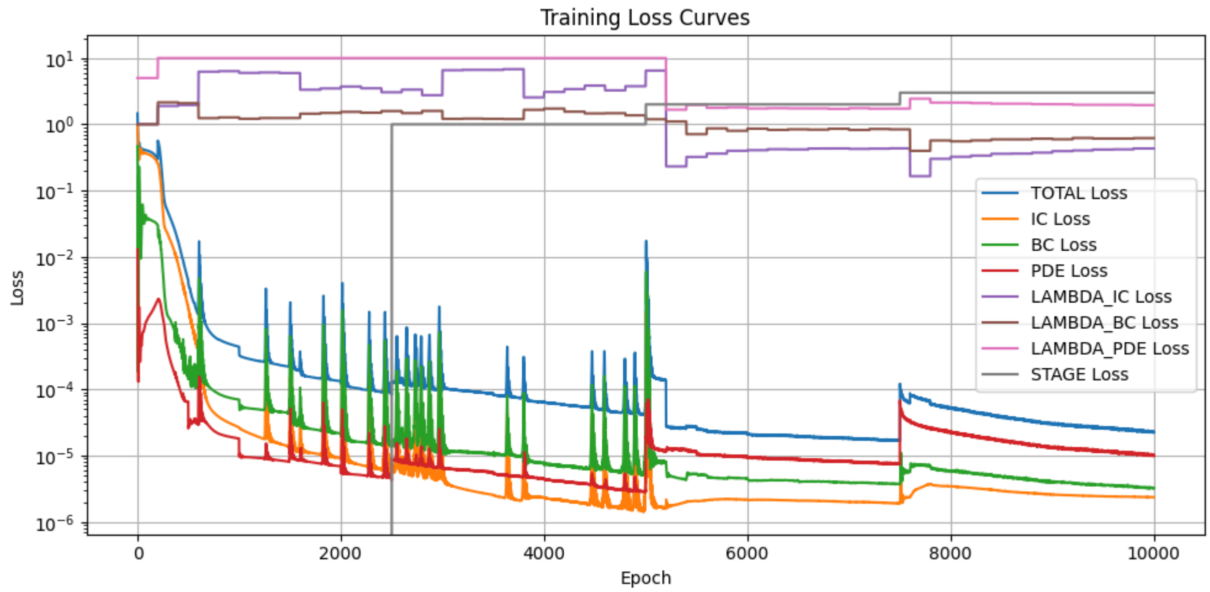
In [6]:
```
ac_clean = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strat
h_ac = ac_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='Heat 
save_training_plots(h_ac, FIGURES+'acpinn_clean_training.png', 'Heat | AC-PINN Clea
ac_clean.plot_loss_history(h_ac)
ac_clean.plot_weight_history(h_ac)
ac_clean.plot_solution(title='AC-PINN - Clean Dense')
save_history(h_ac, RESULTS+'ac_clean_history.npy')
```

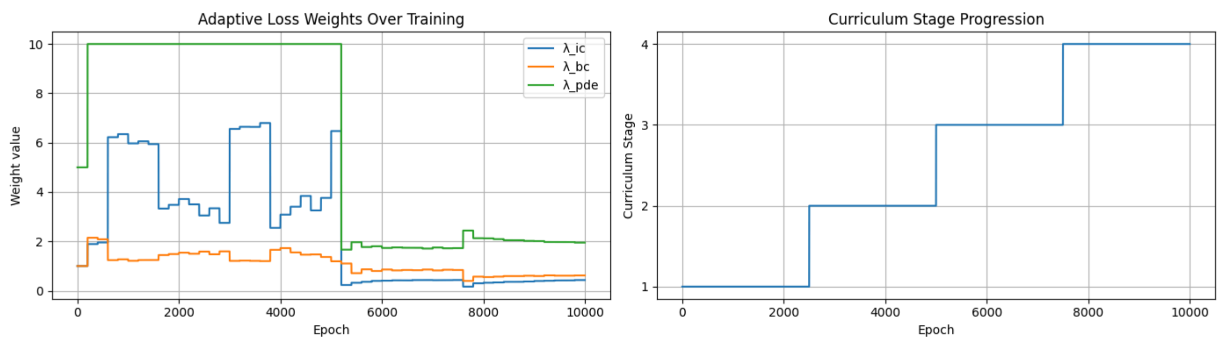
```
[Heat | AC-PINN Clean | Epoch     0/10000] Stage 1/4 | Total: 1.457619 | IC: 0.92624
3 | BC: 0.466691 | PDE: 0.012937 | λ=(1.00,1.00,5.00)
```

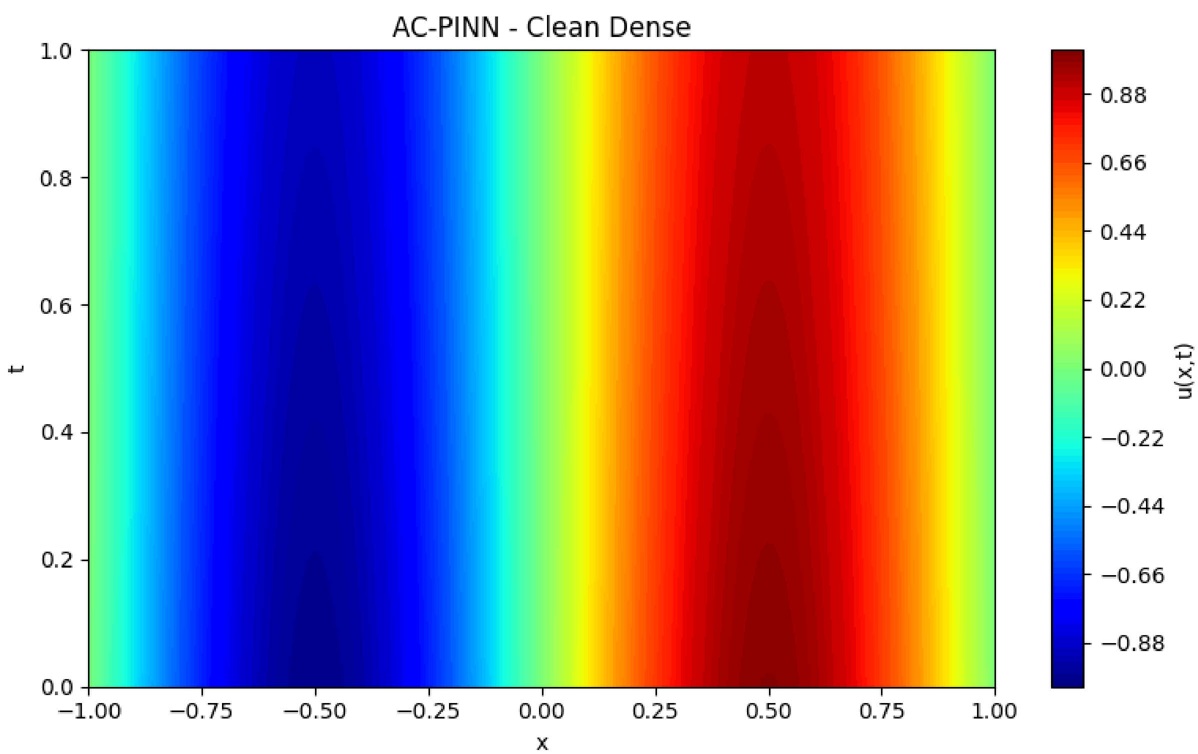
```
[Heat | AC-PINN Clean | Epoch  1000/10000] Stage 1/4 | Total: 0.000341 | IC: 0.00002
7 | BC: 0.000070 | PDE: 0.000010 | λ=(5.97,1.21,10.00)
[Heat | AC-PINN Clean | Epoch  2000/10000] Stage 1/4 | Total: 0.000122 | IC: 0.00000
9 | BC: 0.000022 | PDE: 0.000005 | λ=(3.72,1.54,10.00)
[Heat | AC-PINN Clean | Epoch  3000/10000] Stage 2/4 | Total: 0.000174 | IC: 0.00000
8 | BC: 0.000041 | PDE: 0.000007 | λ=(6.56,1.21,10.00)
[Heat | AC-PINN Clean | Epoch  4000/10000] Stage 2/4 | Total: 0.000064 | IC: 0.00000
2 | BC: 0.000008 | PDE: 0.000004 | λ=(3.08,1.73,10.00)
[Heat | AC-PINN Clean | Epoch  5000/10000] Stage 2/4 | Total: 0.000108 | IC: 0.00000
6 | BC: 0.000036 | PDE: 0.000003 | λ=(6.47,1.19,10.00)
[Heat | AC-PINN Clean | Epoch  6000/10000] Stage 3/4 | Total: 0.000021 | IC: 0.00000
2 | BC: 0.000005 | PDE: 0.000009 | λ=(0.41,0.86,1.73)
[Heat | AC-PINN Clean | Epoch  7000/10000] Stage 3/4 | Total: 0.000019 | IC: 0.00000
2 | BC: 0.000004 | PDE: 0.000008 | λ=(0.43,0.82,1.75)
[Heat | AC-PINN Clean | Epoch  8000/10000] Stage 4/4 | Total: 0.000052 | IC: 0.00000
3 | BC: 0.000006 | PDE: 0.000023 | λ=(0.32,0.55,2.12)
[Heat | AC-PINN Clean | Epoch  9000/10000] Stage 4/4 | Total: 0.000030 | IC: 0.00000
3 | BC: 0.000004 | PDE: 0.000013 | λ=(0.39,0.59,2.01)
AC-PINN training complete in 107.73s
```




```
Saved: ../figures/heat/acpinn_clean_training.png
Saved: ../figures/heat/acpinn_clean_training.csv
```





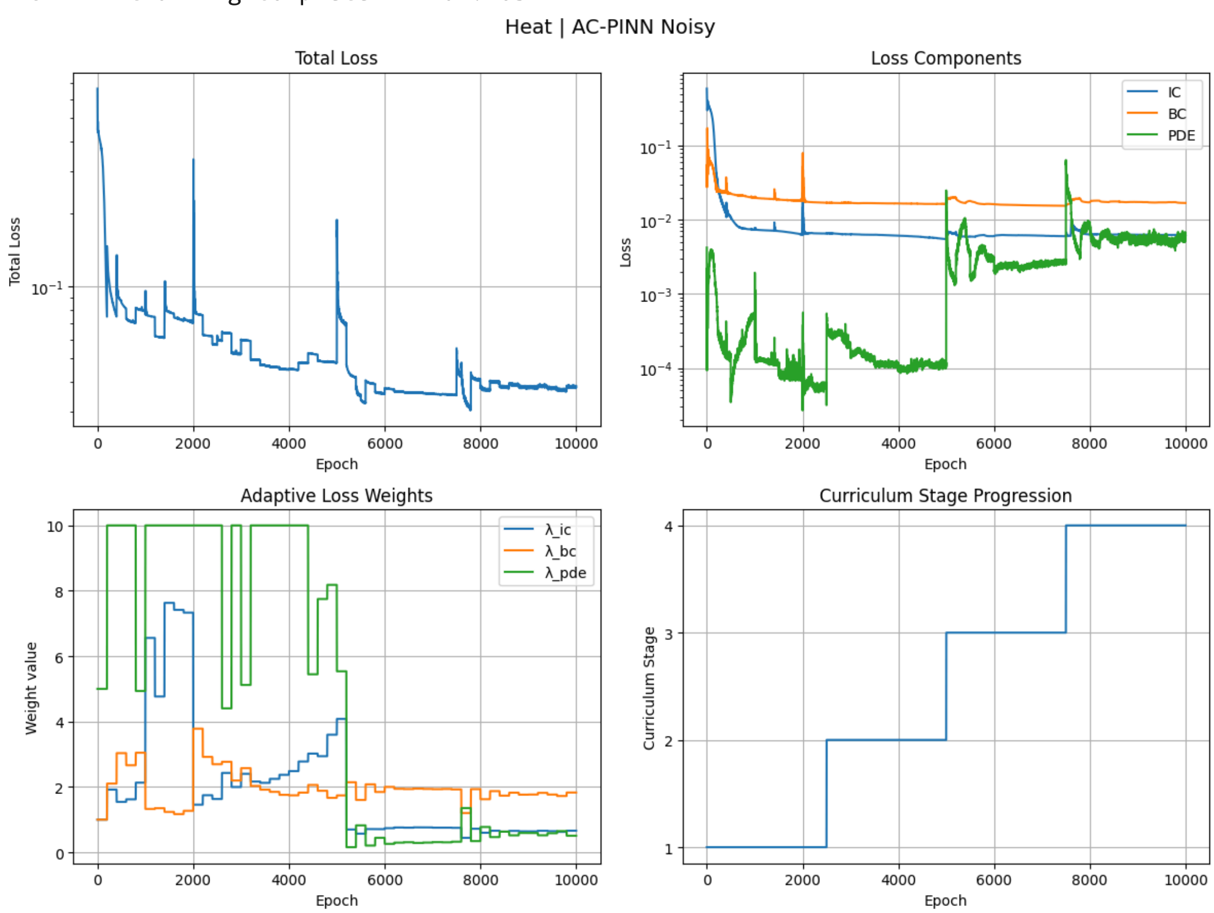



```
Saved: ../results/heat/ac_clean_history.npy
```

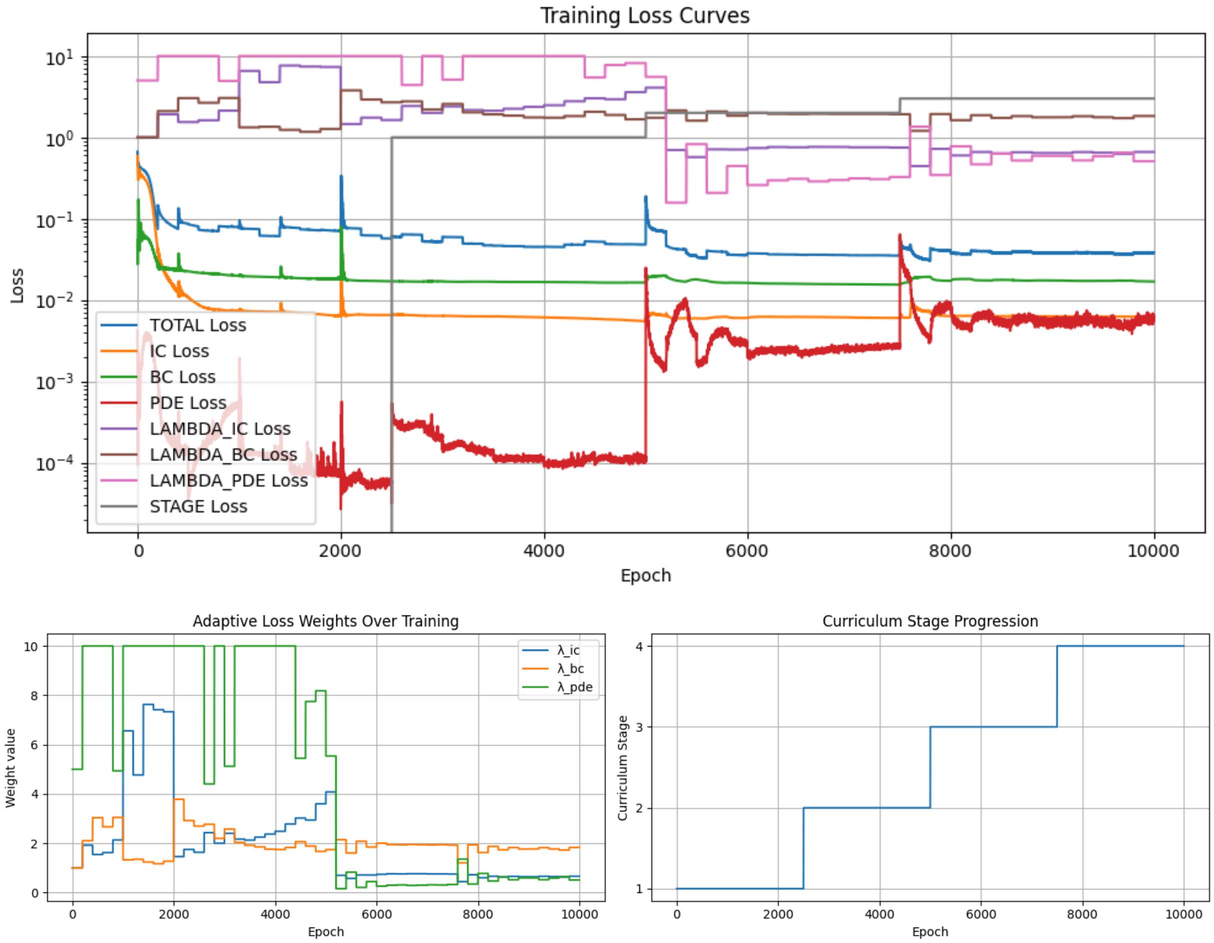
Section 6 - AC-PINN, Noisy Sparse

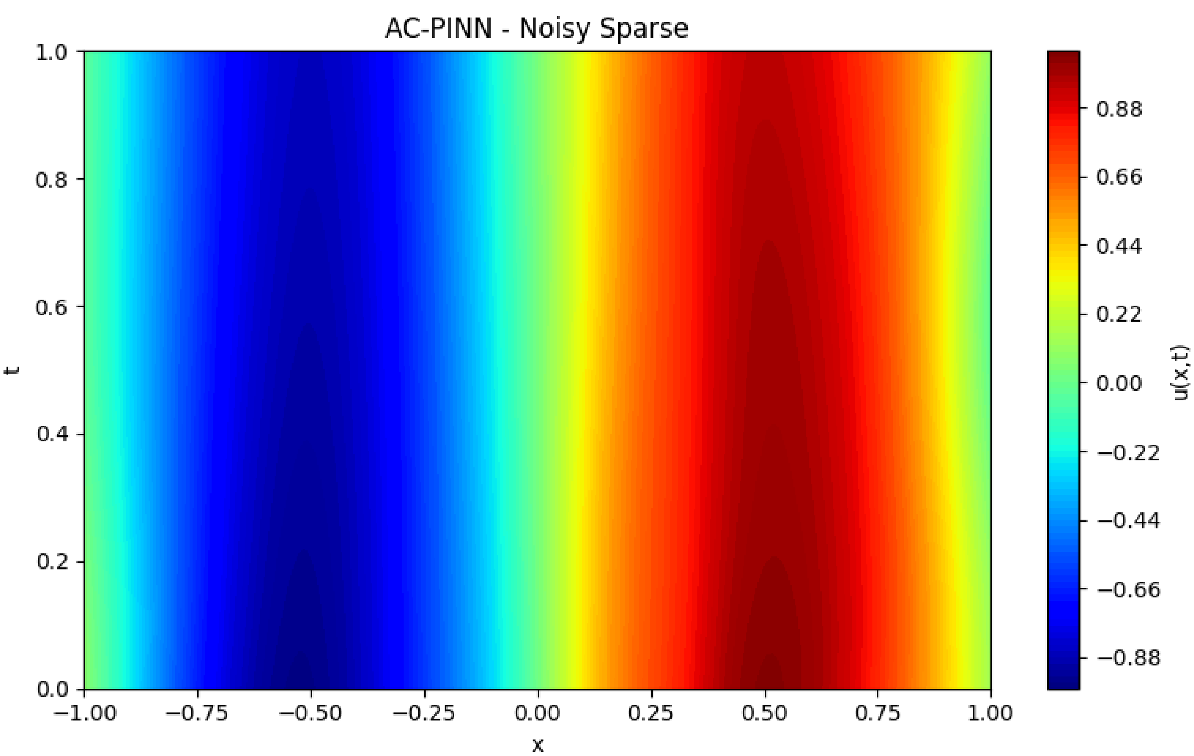
In [7]:
```
ac_noisy = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strat
h_an = ac_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label='Heat
save_training_plots(h_an, FIGURES+'acpinn_noisy_training.png', 'Heat | AC-PINN Nois
ac_noisy.plot_loss_history(h_an)
ac_noisy.plot_weight_history(h_an)
ac_noisy.plot_solution(title='AC-PINN - Noisy Sparse')
save_history(h_an, RESULTS+'ac_noisy_history.npy')
```

```
[Heat | AC-PINN Noisy | Epoch     0/10000] Stage 1/4 | Total: 0.664383 | IC: 0.58908
7 | BC: 0.054160 | PDE: 0.004227 | λ=(1.00,1.00,5.00)
[Heat | AC-PINN Noisy | Epoch  1000/10000] Stage 1/4 | Total: 0.076296 | IC: 0.00739
9 | BC: 0.019922 | PDE: 0.000134 | λ=(6.56,1.33,10.00)
[Heat | AC-PINN Noisy | Epoch  2000/10000] Stage 1/4 | Total: 0.078084 | IC: 0.00635
9 | BC: 0.018122 | PDE: 0.000027 | λ=(1.46,3.78,10.00)
[Heat | AC-PINN Noisy | Epoch  3000/10000] Stage 2/4 | Total: 0.060000 | IC: 0.00638
3 | BC: 0.017043 | PDE: 0.000154 | λ=(2.40,2.57,5.12)
[Heat | AC-PINN Noisy | Epoch  4000/10000] Stage 2/4 | Total: 0.045182 | IC: 0.00612
5 | BC: 0.016669 | PDE: 0.000090 | λ=(2.48,1.74,10.00)
[Heat | AC-PINN Noisy | Epoch  5000/10000] Stage 2/4 | Total: 0.051591 | IC: 0.00548
8 | BC: 0.016443 | PDE: 0.000103 | λ=(4.08,1.74,5.54)
[Heat | AC-PINN Noisy | Epoch  6000/10000] Stage 3/4 | Total: 0.037621 | IC: 0.00602
1 | BC: 0.016299 | PDE: 0.002111 | λ=(0.74,2.00,0.26)
[Heat | AC-PINN Noisy | Epoch  7000/10000] Stage 3/4 | Total: 0.035607 | IC: 0.00610
7 | BC: 0.015631 | PDE: 0.002530 | λ=(0.75,1.93,0.31)
[Heat | AC-PINN Noisy | Epoch  8000/10000] Stage 4/4 | Total: 0.038081 | IC: 0.00634
1 | BC: 0.017183 | PDE: 0.008197 | λ=(0.60,1.63,0.78)
[Heat | AC-PINN Noisy | Epoch  9000/10000] Stage 4/4 | Total: 0.038238 | IC: 0.00630
4 | BC: 0.017397 | PDE: 0.005816 | λ=(0.64,1.77,0.59)
AC-PINN training complete in 102.46s
```



```
Saved: ../figures/heat/acpinn_noisy_training.png
Saved: ../figures/heat/acpinn_noisy_training.csv
```



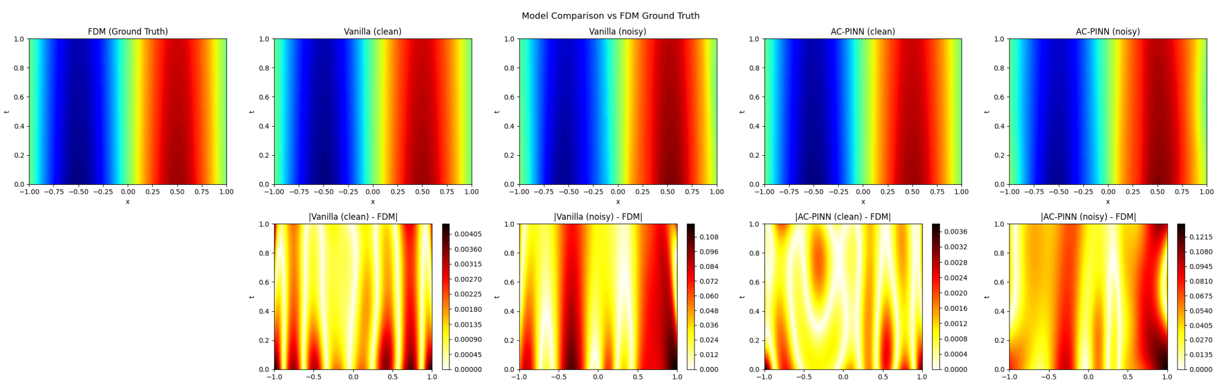


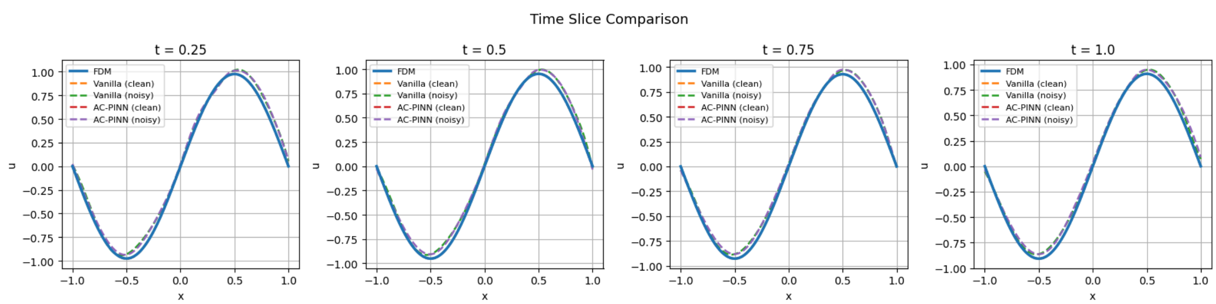

```
Saved: ../results/heat/ac_noisy_history.npy
```

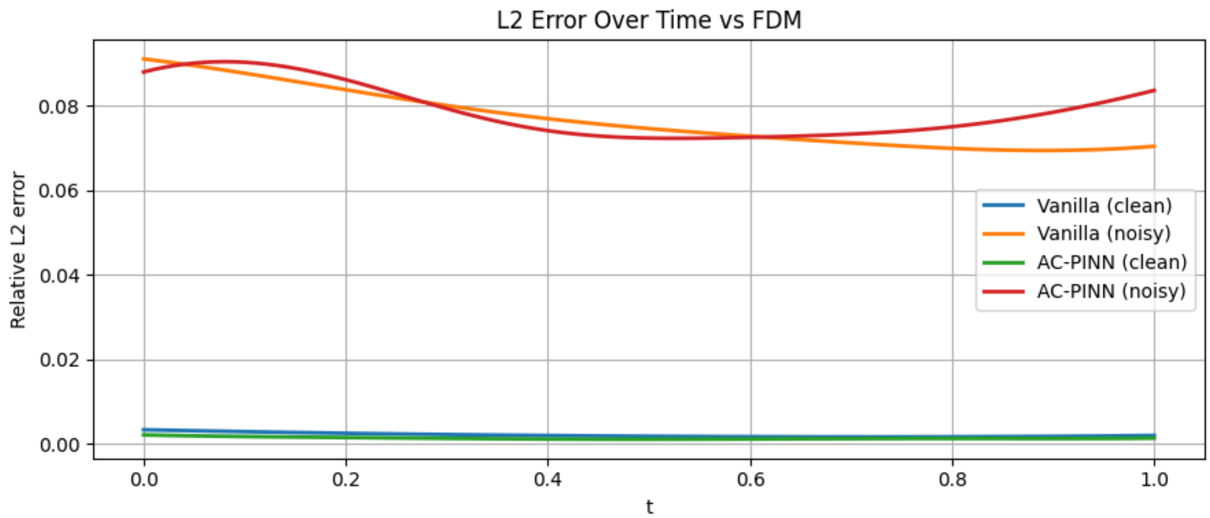
Section 7 - Benchmark vs FDM

In [8]:
```python
bench = Benchmark(fdm, nx=200, nt=100)
bench.add('Vanilla (clean)', vanilla_clean)
bench.add('Vanilla (noisy)', vanilla_noisy)
bench.add('AC-PINN (clean)', ac_clean)
bench.add('AC-PINN (noisy)', ac_noisy)
bench.run()

metrics = bench.compare_metrics()
bench.plot_comparison(save_path=FIGURES+'comparison.png')
bench.plot_time_slices(save_path=FIGURES+'time_slices.png')
bench.plot_error_over_time(save_path=FIGURES+'error_over_time.png')

save_metrics(metrics, RESULTS+'benchmark_metrics.npy')
print('Heat experiments complete.')
```

```
=================================================================
  Model                          Rel L2     Max Err        MAE        RMSE
=================================================================
  Vanilla (clean)              0.002163   0.004318   0.001199   0.001454
  Vanilla (noisy)              0.077401   0.117756   0.043770   0.052030
  AC-PINN (clean)              0.001364   0.003767   0.000739   0.000917
  AC-PINN (noisy)              0.079294   0.132765   0.046997   0.053303
=================================================================
```








```
Saved: ../results/heat/benchmark_metrics.npy
Heat experiments complete.
```

Section 8 - Noise Level Study

In [9]:

```python
noise_results = {}
for eps in [0.05, 0.1, 0.2]:
    print(f'\n--- ε={eps} ---')
    d = gen.generate(N_ic=50, N_bc=50, N_f=3000, noise_eps=eps)
    v = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
    v.fit(d, epochs=5000, print_every=2500)
    a = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strategy
    a.fit(d, epochs=5000, print_every=2500)
    b = Benchmark(fdm).add(f'Vanilla ε={eps}', v).add(f'AC-PINN ε={eps}', a)
    b.run()
    noise_results[eps] = b.compare_metrics()
save_metrics(noise_results, RESULTS+'noise_study_metrics.npy')
```

```
--- ε=0.05 ---
[Epoch     0/5000] Total: 1.188090 | IC: 0.338890 | BC: 0.176421 | PDE: 0.134556
[Epoch  2500/5000] Total: 0.005528 | IC: 0.002129 | BC: 0.003306 | PDE: 0.000019
Training complete in 44.03s
[Epoch     0/5000] Stage 1/4 | Total: 1.035548 | IC: 0.344078 | BC: 0.211931 | PDE:
0.095908 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.014728 | IC: 0.002124 | BC: 0.003371 | PDE:
0.000016 | λ=(4.75,1.33,10.00)
AC-PINN training complete in 50.39s
=================================================================
  Model                         Rel L2    Max Err       MAE       RMSE
=================================================================
  Vanilla ε=0.05              0.012301   0.031044   0.007165   0.008269
  AC-PINN ε=0.05              0.015576   0.039822   0.008610   0.010470
=================================================================

--- ε=0.1 ---
[Epoch     0/5000] Total: 1.000255 | IC: 0.244167 | BC: 0.555901 | PDE: 0.040037
[Epoch  2500/5000] Total: 0.029382 | IC: 0.009839 | BC: 0.019307 | PDE: 0.000047
```

```
Training complete in 44.37s
[Epoch     0/5000] Stage 1/4 | Total: 0.621207 | IC: 0.442940 | BC: 0.067016 | PDE:
0.022250 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.067459 | IC: 0.010093 | BC: 0.020429 | PDE:
0.000091 | λ=(2.71,1.92,9.00)
AC-PINN training complete in 52.63s
=================================================================
  Model                         Rel L2    Max Err        MAE       RMSE
=================================================================
  Vanilla ε=0.1               0.067209   0.085943   0.040670   0.045179
  AC-PINN ε=0.1               0.075193   0.092962   0.043413   0.050546
=================================================================

--- ε=0.2 ---
[Epoch     0/5000] Total: 0.973765 | IC: 0.416146 | BC: 0.170493 | PDE: 0.077425
[Epoch  2500/5000] Total: 0.123071 | IC: 0.022768 | BC: 0.099331 | PDE: 0.000195
Training complete in 45.13s
[Epoch     0/5000] Stage 1/4 | Total: 2.398360 | IC: 1.232614 | BC: 1.159781 | PDE:
0.001193 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.255087 | IC: 0.020972 | BC: 0.093955 | PDE:
0.000035 | λ=(6.63,1.23,10.00)
AC-PINN training complete in 52.10s
=================================================================
  Model                         Rel L2    Max Err        MAE       RMSE
=================================================================
  Vanilla ε=0.2               0.060566   0.121726   0.031003   0.040714
  AC-PINN ε=0.2               0.069335   0.134233   0.036272   0.046608
=================================================================
Saved: ../results/heat/noise_study_metrics.npy
```